

Unmasking Bots in Higher Dimensions: Message Passing over Simplexes for Bot Detection

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Abstract

Detecting social bots is critical to ensuring the security of online discourse and maintaining trust in social networks. Early feature-based and text-based methods often fail against bots that mimic human behavior, and graph-based approaches have emerged to better exploit structural signals. However, most existing Graph Neural Networks (GNNs) still focus on pairwise connections, overlooking higher-order relational patterns, and their multi-relation fusion strategies are typically simplistic, ignoring dependencies between relations and user-specific preferences. To overcome these limitations, we propose MPS-Bot, a model that integrates higher-order structure modeling with user-specific cross-relation dependency learning. MPS-Bot introduces a simplex convolutional layer that leverages simplexes derived from network structures to capture group coordination patterns beyond pairwise connections. In addition, a cross-relation dependency attention mechanism adaptively fuses relation-specific representations according to each user's relational preferences, leading to more discriminative and robust multi-relation representations. Extensive experiments on two widely used Twitter bot detection benchmarks, MGTAB and TwiBot-22, show that MPS-Bot generally outperforms state-of-the-art baselines. These findings highlight the effectiveness of higher-dimensional message passing over simplexes as a powerful approach to unmasking bots in social networks.

CCS Concepts

- Security and privacy → Social network security and privacy;
- Computing methodologies → Machine learning.

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Keywords

Social Bot Detection, Simplicial Complex, Graph Neural Networks

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1 Introduction

As social media becomes a central channel for information sharing and public discussion, growing attention has turned to the rising presence of social bots. These bots are increasingly active across online platforms, posing serious challenges to the quality of digital communication. By spreading misinformation [39, 43, 48], manipulating public opinion [6, 13], and interfering with political events [15, 19, 33], they undermine user trust and weaken the authenticity of online interactions [2, 34]. As a result, developing reliable and effective detection methods has become a key research focus, and is essential for maintaining trust and authenticity on the Web.

Early detection approaches largely relied on feature-based methods, which use engineered attributes such as user metadata [22] in combination with traditional classifiers. While effective on early benchmarks, these models often fail against bots that deliberately mimic human traits. Text-based methods emerged, applying text analysis techniques such as language models [8, 26] to analyze textual information, aim to detect malicious discourse or abnormal topic patterns. However, when bots fabricate human-like text [5, 12, 29], these approaches also lose effectiveness. In recent years, graph-based methods have become as a dominant paradigm for social bot detection. By interpreting users as nodes and their interactions as edges, these approaches capture relational structures and leverage Graph Neural Networks (GNNs) [1] to learn expressive representations. For example, RGT [9] enhances detection by encoding heterogeneous social graphs through relation-aware mechanisms, while BotHP [16] further exploits both homophilic and heterophilic

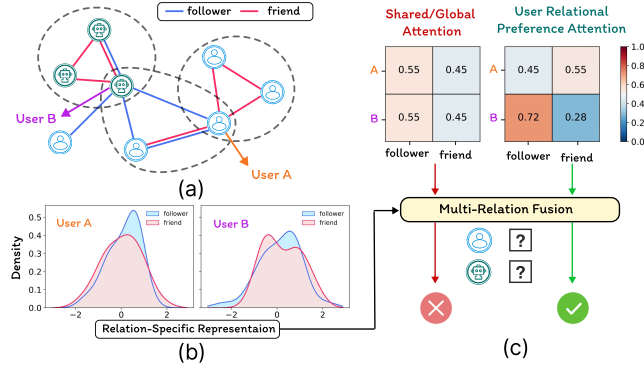


Figure 1: An illustration of two challenges: capturing higher-order interactions and improving simple relation fusion.

edge patterns via generative graph self-supervision, thereby improving detection performance.

Despite significant advances, current social bot detectors still struggle with two fundamental issues: (1) Overlooked higher-order structures. Users in social networks such as the Twitter network often interact in groups that go beyond pairwise connections [7, 10, 28, 45], as shown in Figure 1 (a) by the black dashed circles. However, most GNN-based methods restrict message passing to edges or limited neighborhoods, making it difficult to capture such higher-order coordination patterns and causing them to miss important group-level signals [4]. (2) Limitations of conventional multi-relation fusion. In multi-relational social graphs, links differ in semantic meaning. This diversity drives users to exhibit varying preferences across relations. As shown by the KDE distributions in Figure 1(b), different users (e.g., User A and User B) learn distinct relation-specific representations, reflecting these preferences. However, most existing models overlook such user-specific relational diversity. They process each relation independently and then merge the outputs using simple operations such as summation [11], averaging [14], concatenation [27, 49, 50], or shared attention weights [9, 42] that assign the same importance weights to all users, as illustrated in the upper-left part of Figure 1(c). Such fusion diminishes the model’s discriminative capacity, as shown by the detection result highlighted by the red arrow in Figure 1(c).

To address these challenges, we propose MPS-Bot (Message Passing over Simplexes for Bot Detection), a novel model that integrates higher-order structure modeling with user-specific cross-relation dependency learning. Specifically, MPS-Bot leverages simplicial complexes that represent higher-order relations beyond pairwise connections, to design a simplex convolutional layer, which enables message passing over simplexes, thereby capturing the coordination patterns of user groups. For relation-specific representations learned from simplex convolution, MPS-Bot aligns their fusion with each user’s relational preferences, as illustrated in the upper-right part of Figure 1(c). This design facilitates more effective multi-relation fusion and consequently enhances the model’s discriminative power for social bot detection, as indicated by the green arrow in Figure 1(c). The main contributions of our work are summarized:

- (1) **Higher-order structure modeling.** We design a simplex convolutional layer based on simplicial complexes to exploit simplexes, enabling message passing between nodes

and higher-order structures to capture group coordination patterns that pairwise methods overlook.

- (2) **User-specific cross-relation dependency modeling.** We introduce a relation-level dependency-aware attention mechanism that models interactions across relations while adapting the fusion to each user’s relational preferences, resulting in more effective and discriminative multi-relation fusion.
- (3) **Comprehensive experimental evaluation.** We conduct extensive experiments on two widely used twitter bot detection benchmarks, showing that MPS-Bot consistently outperforms state-of-the-art methods and validating the benefits of combining higher-order structure with user-specific cross-relation dependency modeling.

2 Related Works

2.1 Graph-based Bot Detection

Social bot detection aims to distinguish automated accounts from genuine users in online platforms. Recently, graph-based methods [20, 25] have become a paradigm due to their ability to model the structure of social networks [10]. These approaches represent users as nodes and interactions as edges, and leverage Graph Neural Networks (GNNs) to learn node embeddings by aggregating information from their neighbors. BotRGCN [11] represents multi-modal user information as a heterogeneous graph and applies relational GCNs, while RGT [9] leverages heterogeneous information networks with relational graph transformers and semantic attention. Building on this foundation, SEBot [47] constructs multi-view graphs and integrates structural entropy minimization with heterophily-aware contrastive learning. CACL [3] introduces a community-aware module with graph contrastive learning to mine hard samples. BotHP [16] develops a generative graph self-supervised framework that unifies heterophily-aware representation learning with prototype-guided clustering. However, many graph-based methods model only pairwise relations, neglecting higher-order structures common in coordinated user groups.

To address this gap, some studies have explored high-order modeling through topological techniques. The work [32] employs simplicial complexes with persistent homology to extract higher-order features from each user’s ego network, but these features are pre-computed and static, preventing end-to-end optimization with the detection task and limiting adaptability. Another direction, exemplified by HHG-Bot [41], integrates heterogeneous graph convolutions with hypergraph aggregation to capture coarse-grained multi-node interactions. While this approach models high-order connectivity, it does not explicitly encode motif-level patterns and still relies on relatively coarse relation fusion, potentially missing fine-grained cross-relation dependencies. As a result, existing higher-order methods either lack adaptive, trainable representations or fail to capture detailed structural cues, leaving room for more expressive and robust modeling strategies.

2.2 Relational Modeling in Social Bot Detection

Social networks are inherently multi-relational [35], where each edge type carries distinct semantics and varying degrees of reliability. Effectively modeling these heterogeneous relations is critical for bot detection.

To handle relational diversity, several approaches [11, 14, 27, 49, 50] treat each relation type separately and then merge the outputs using simple fusion operations such as summation [11], averaging [14], or concatenation [27, 49, 50]. While these methods capture relation-specific information, their fusion strategies treat all relations uniformly, preventing the model from distinguishing valuable relations from less reliable ones. Moreover, fixed fusion schemes limit the ability to adjust relation importance based on structural or semantic context. These limitations motivate the development of attention-based fusion mechanisms for more flexible and adaptive multi-relational modeling.

To address these limitations, other studies [9, 42] employ attention mechanisms to capture cross-relation dependencies. For instance, RGT [9] applies a shared attention mechanism across relations, using the same set of relation-level attention parameters for all nodes when fusing multi-relational information. Similarly, BotRGA [42] computes a global relation weight by averaging the attention scores of all nodes under each specific relation, resulting in a uniform relation weight shared across the entire graph. Consequently, the attention parameters are globally shared, hindering adaptation to user-specific relational preferences.

These drawbacks highlight the need for adaptive fusion mechanisms that model user-specific cross-relation dependencies.

3 Preliminaries

In this section, we present some preliminary knowledge.

3.1 Problem Definition

We formulate social bot detection as a semi-supervised node classification problem on a multi-relational social graph. Formally, we construct a graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathbf{X}, \mathcal{Y}\}$, where $\mathcal{V} = \{v_i \mid i = 1, 2, \dots, n\}$ denotes the set of user nodes, $\mathcal{E} = \bigcup_{r=1}^R \mathcal{E}_r$ is the set of edges formed by R different relation types (e.g., “follower” and “following” links), and $\mathbf{X}$ represents the feature matrix. $\mathcal{Y} = \{y_i\}$ denotes the label set, where $y_i = 0$ represents a human user and $y_i = 1$ represents a bot user. Given the graph $\mathcal{G}$ and the labels of training nodes $\mathcal{Y}_{\text{train}}$, the objective is to learn a function:

$$f(\mathcal{G}, \mathcal{Y}_{\text{train}}) \rightarrow \hat{\mathcal{Y}}_{\text{test}}, \quad (1)$$

which predicts the labels of the testing nodes $\mathcal{Y}_{\text{test}}$.

3.2 Simplicial Complex

Simplicial complexes are capable of capturing higher-order interactions among two or more nodes that are represented by simplexes [23, 37, 38, 44, 46]. Formally, a k -simplex is a set of $k + 1$ nodes representing a k -dimensional fully connected structure. Specifically, a 0-simplex is a node, a 1-simplex is an edge, and a 2-simplex is a triangle formed by three mutually connected nodes.

In the context of social networks, simplicial complexes naturally capture higher-order user interactions that go beyond pairwise connections. For instance, a 2-simplex can be used to model a triadic interaction, representing a group of three users who are mutually connected. If these users are denoted by v_i, v_j , and v_k , their interaction forms a 2-simplex, which can be represented as $\tau = (v_i, v_j, v_k)$.

By incorporating simplicial complexes into social bot representation learning, information can flow through simplexes, enabling

the model to capture higher-order structural signals that are critical for identifying coordinated user behaviors.

4 Methodology

4.1 Overview

Figure 2 illustrates our proposed model, which comprises three main components. First, we model the real-world social network as a multi-relational graph, where users are nodes and different interaction types define distinct relations. User features are initialized with the dataset-provided attributes. Second, we introduce the HeteroSimplexLayer, which integrates higher-order structure modeling and adaptive relation fusion. For each relation type, we extract simplexes and apply the proposed simplex convolutional layer to perform message passing on these simplexes, thereby learning relation-specific representations. These representations are then fused through a user-specific cross-relation dependency attention mechanism, which models interactions across relations while adapting the fusion to each user’s relational preferences. Finally, we stack L HeteroSimplexLayers to form a hierarchical model that refines node representations, and input the final representations into a classifier to distinguish human users from bots.

4.2 Message Passing over Simplexes

Real-world social networks exhibit higher-order interactions, since users tend to interact in groups rather than in pairs. To capture such group coordination patterns, we extend the original multi-relational social graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathbf{X}, \mathcal{Y}\}$ by explicitly extracting 2-simplexes from each relation r .

Formally, for a relation-specific edge set $\mathcal{E}_r$, a 2-simplex is denoted by (v_i, v_j, v_k) if all three pairwise edges exist. We enumerate such simplexes and obtain $\mathcal{T}_r$ for each relation r :

$$\mathcal{T}_r = \{(v_i, v_j, v_k) \mid (v_i, v_j), (v_j, v_k), (v_i, v_k) \in \mathcal{E}_r\}. \quad (2)$$

This process produces relation-wise simplexes, providing the structural basis for the subsequent message passing module.

Next, we introduce the message passing component of a HeteroSimplexLayer, which operates on the relation-wise simplexes extracted above and produces relation-specific node representations.

Most existing GNN-based methods only exchange messages along edges, limiting their ability to capture coordination patterns among users. To address this limitation, we introduce *SimplexConv*, a simplex convolutional layer that propagates messages over simplexes, allowing nodes to exchange information through shared simplexes. The following describes how *SimplexConv* conducts message passing through the extracted simplexes.

Given node feature matrix $\mathbf{X}$, we denote the feature of node v_i as x_i under the row-vector convention. Let r denote a specific relation type, and let $\mathcal{T}_r$ be the set of 2-simplexes extracted under relation r . For each relation-specific simplex $\tau = (v_i, v_j, v_k) \in \mathcal{T}_r$, we first transform the incident node features with a fully connected layer parameterized by learnable weights $\mathbf{W}_n$, and then derive a latent simplex representation by averaging the transformed features, where $|\tau|$ denotes the number of nodes in simplex τ :

$$z_\tau^r = \frac{1}{|\tau|} \sum_{v \in \tau} x_v \mathbf{W}_n \quad (3)$$

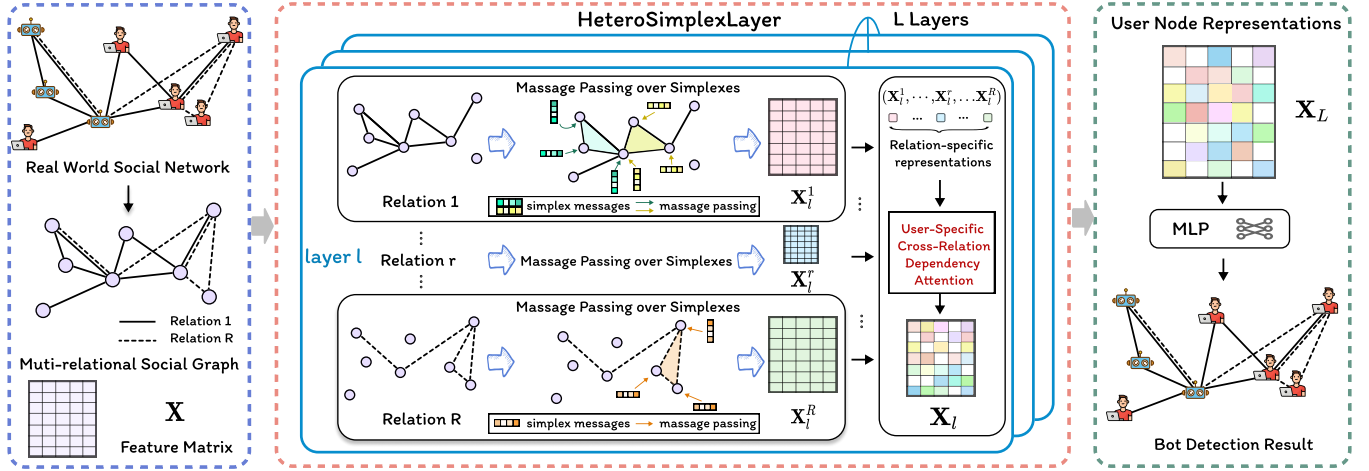


Figure 2: The framework of our proposed MPS-Bot model.

This representation is further mapped by another fully connected layer with weights $\mathbf{W}_l$ to produce a simplex message:

$$m_\tau^r = z_\tau^r \mathbf{W}_l \quad (4)$$

The message is then scattered back to their incident nodes through mean aggregation. Specifically, let $\mathcal{T}_r(i) = \{\tau \in \mathcal{T}_r \mid v_i \in \tau\}$ denote the set of simplexes containing node v_i , the node update is

$$x_i^r = \frac{1}{|\mathcal{T}_r(i)|} \sum_{\tau \in \mathcal{T}_r(i)} m_\tau^r \quad (5)$$

If a node does not belong to any 2-simplex under relation r , i.e., $|\mathcal{T}_r(i)| = 0$, we skip 2-simplex-based aggregation and instead apply a linear transformation followed by nonlinearity and dropout directly to its feature vector.

This message passing over simplexes allows information to be exchanged through shared 2-simplexes instead of only along edges. Consequently, the model captures higher-order structures that underlie coordination patterns among users.

4.3 User-Specific Cross-Relation Dependency Modeling

After the message passing stage, each relation r produces a relation-specific representation, denoted as $\mathbf{X}_l^r$ at the l -th layer. These representations are obtained through gated updates that combine the original node features with relation-specific signals. Although each $\mathbf{X}_l^r$ encodes structural information under its own relation type, the resulting representations are learned in isolation, without modeling dependencies across relations and user-specific preferences.

To address these limitations, we introduce a user-specific cross-relation dependency attention mechanism, inspired by the Transformer architecture [40], which explicitly captures how relations interact each other for each user. We stack the relation-specific representations along the relation dimension as

$$\mathbf{X}_l = [\mathbf{X}_l^1; \dots; \mathbf{X}_l^R], \quad (6)$$

where the second dimension corresponds to relation types. For each node v_i , its relation-specific representations $\{\mathbf{x}_{i,l}^1, \dots, \mathbf{x}_{i,l}^R\}$ are treated as R relation-wise features, and attention is applied across relations to capture their dependencies.

The attention-enhanced representation with respect to relation r is computed as:

$$\tilde{\mathbf{x}}_{i,l}^r = \sum_{s=1}^R \alpha_{rs} \mathbf{x}_{i,l}^s \mathbf{W}_V, \quad (7)$$

where the attention weight α_{rs} is given by

$$\alpha_{rs} = \frac{\exp((\mathbf{x}_{i,l}^r \mathbf{W}_Q)(\mathbf{x}_{i,l}^s \mathbf{W}_K)^\top / \sqrt{d})}{\sum_{t=1}^R \exp((\mathbf{x}_{i,l}^r \mathbf{W}_Q)(\mathbf{x}_{i,l}^t \mathbf{W}_K)^\top / \sqrt{d})}, \quad (8)$$

and d denotes the feature dimension of the relation-specific representations. Here, $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$ are learnable projection matrices. $\tilde{\mathbf{x}}_{i,l}^r$ represents the relation- r representation of node v_i after attending to all its relation-specific representations $\{\mathbf{x}_{i,l}^1, \dots, \mathbf{x}_{i,l}^R\}$. Note that the index s in the summation simply enumerates all relations and does not introduce a new symbol.

Finally, the attended relation representations are aggregated by mean pooling:

$$\mathbf{x}_i^{(l)} = \frac{1}{R} \sum_{r=1}^R \tilde{\mathbf{x}}_{i,l}^r, \quad (9)$$

which yields the fused representation for node v_i at layer l . Collectively, the fused feature matrix is denoted as $\mathbf{X}_l$.

In summary, this user-specific cross-relation dependency attention mechanism enables each user to integrate information from multiple relation types in a personalized and coordinated manner. By modeling attention along the relation dimension, it captures both inter-relation interactions and user-specific relational preferences, resulting in fused node representations that are more expressive and better suited for bot detection.

4.4 Hierarchical Representation Learning

Having defined the two components of a HeteroSimplexLayer, we stack L such layers to form a hierarchical model that captures increasingly complex patterns underlying group coordination. Let $\mathbf{X}_0$ denote the initial node feature matrix, i.e., the input representation before iterative updates:

$$\begin{aligned} \mathbf{X}_0 &= \mathbf{X}, \\ \mathbf{X}_l &= \text{HeteroSimplexLayer}^{(l)}(\mathbf{X}_{l-1}), \quad l = 1, \dots, L, \end{aligned} \quad (10)$$

Algorithm 1 MPS-Bot

Input: Multi-relational social graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathbf{X}, \mathcal{Y}\}$, number of layers L

Output: Predicted node labels $\hat{y}_i$ for $v_i \in \mathcal{V}$

- 1: Extract relation-wise 2-simplexes $\{\mathcal{T}_r\}_{r=1}^R$ from $\mathcal{G}$
- 2: Initialize node features:

$$\mathbf{X}_0 = \mathbf{X}$$

- 3: **for** $l = 1$ **to** L **do**
- 4: **for** each relation $r = 1, \dots, R$ **do**
- 5: Compute simplex messages from $\mathcal{T}_r$ via Eq. (4).
- 6: Perform message passing to obtain relation-specific node representations via Eq. (5) to obtain $\mathbf{X}_l^r = \{\mathbf{x}_i^r\}_{i=1}^{|\mathcal{V}|}$.
- 7: **end for**
- 8: Fuse $\{\mathbf{X}_l^r\}_{r=1}^R$ with user-specific cross-relation dependency attention via Eq. (7) and Eq. (9):
- 9: Collect fused representations:

$$\mathbf{X}_l = \{\mathbf{x}_i^{(l)}\}_{i=1}^{|\mathcal{V}|}$$

10: **end for**

11: Predict node labels with classifier:

$$\hat{y}_i = \text{softmax}(\phi(\mathbf{x}_i^{(L)} \mathbf{W}_1 + b_1) \mathbf{W}_2 + b_2)$$

where $\mathbf{X}_l$ denotes the fused node feature matrix output by the l -th HeteroSimplexLayer, and $\mathbf{x}_i^{(l)}$ is the representation of node v_i corresponding to the i -th row of $\mathbf{X}_l$.

At the top layer, the final node representations $\mathbf{X}_L$ are fed into a MLP classifier:

$$\hat{y}_i = \text{softmax}(\phi(\mathbf{x}_i^{(L)} \mathbf{W}_1 + b_1) \mathbf{W}_2 + b_2), \quad (11)$$

where $\hat{y}_i$ denotes the predicted label distribution of node v_i , ϕ is a non-linear activation followed by dropout, and $\mathbf{W}_1, \mathbf{W}_2, b_1, b_2$ are learnable parameters of the classifier. The model is trained in an end-to-end manner by minimizing the cross-entropy loss over labeled nodes:

$$\mathcal{L} = -\frac{1}{|\mathcal{V}_{\text{train}}|} \sum_{v_i \in \mathcal{V}_{\text{train}}} y_i \log \hat{y}_i \quad (12)$$

where y_i denotes the ground-truth label distribution of node v_i and $\mathcal{V}_{\text{train}}$ is the set of training nodes.

This hierarchical architecture enables the model to progressively refine node representations, capturing higher-order structural patterns and adaptively modeling cross-relation interactions through simplex-based message passing and user-specific relation fusion.

4.5 Complexity Analysis

The overall procedure of our model is summarized in Algorithm 1. Let $|\mathcal{V}|$ be the number of nodes, R the number of relation types, $|\mathcal{T}_r|$ the number of 2-simplexes under relation r , C the hidden dimension, and L the number of stacked layers.

The extraction of 2-simplexes (line 1) is carried out once as pre-processing. For relation r , this requires intersecting neighbor sets with complexity $\mathcal{O}(\sum_{v \in \mathcal{V}} (d_v^r)^2)$ [36], where d_v^r denotes the degree of node v under relation r . Since this step is offline, it does not affect the training complexity. During training, the main cost of each

Table 1: Hyperparameter setting on MGTAB and Twibot-22.

Parameter	MGTAB	Twibot-22	Parameter	MGTAB	Twibot-22
optimizer	Adam	Adam	hidden dimension	32	128
learning rate	1e-3	1e-3	dropout	0.1	0.5
epoch	200	200	l_2 regularization	3e-5	3e-5

HeteroSimplexLayer comes from two sources. First, *SimplexConv* performs message passing over simplexes (lines 5-6), which is dominated by $\mathcal{O}(|\mathcal{T}_r|C^2)$ per relation. Second, the user-specific cross-relation dependency attention operates along the relation dimension (line 8), with cost $\mathcal{O}(|\mathcal{V}|(R^2C+RC^2))$. Stacking L layers (lines 3-10) yields an overall complexity of $\mathcal{O}(L \cdot (R|\mathcal{T}_r|C^2 + |\mathcal{V}|(R^2C+RC^2)))$. As the number of relations R is typically small and $|\mathcal{T}_r|$ grows roughly linearly with $|\mathcal{E}_r|$ in sparse graphs, the proposed framework remains scalable to large-scale social networks.

5 Experiments

In this section, we perform the extensive experiments in order to evaluate the performance of the proposed model.

5.1 Experimental Preliminaries

5.1.1 Datasets. We provide a brief overview of the social bot detection datasets. 1) **MGTAB** [35] includes 10,199 expert-annotated accounts and seven types of relationships. 2) **Twibot-22** [10] is the largest Twitter bot detection dataset to date, comprising 1,000,000 users with diverse entity types and relationships. These datasets are well-suited for graph-based methods as they are inherently graph-structured. We mainly use follower and friend relationships on MGTAB, and following and followers relationships on Twibot-22. The dataset splits for training, validation, and testing follow the same settings as in previous studies [10, 35]. For Twibot-22, we follow the same user information encoding procedure as in [11] to ensure fairness.

5.1.2 Baselines. To demonstrate the effectiveness of our proposed model MPS-Bot, we select several mainstream social bot detection methods as baselines. These methods are described as follows:

- **GCN** [21] applies a layer-wise propagation rule based on a first-order spectral approximation to aggregate node feature information for semi-supervised classification.
- **BotRGCN** [11] represents multi-modal user information as a heterogeneous graph and employs relational GCNs for social bot detection.
- **RGT** [9] constructs heterogeneous information networks and uses relational graph transformers with semantic attention to obtain node representations.
- **HGT** [20] adopts a transformer architecture for heterogeneous graphs, using relation-specific parameters to model message passing across different node and edge types.
- **SimpleHGN** [30] enhances heterogeneous graph attention networks by incorporating learnable edge embeddings, adding residual connections, and normalizing outputs with L2 constraints.
- **GraphMAE** [18] is a generative self-supervised learning method that learns graph representations by reconstructing

Table 2: Performance comparison on MGTab and Twibot-22 datasets.

Methods	MGTab					Twibot-22				
	Accuracy	F1	Precision	Recall	ROC-AUC	Accuracy	F1	Precision	Recall	ROC-AUC
GCN	85.20±0.87	79.09±2.42	83.82±2.07	77.26±3.99	90.49±0.56	78.28±0.17	52.96±1.53	73.14±1.36	41.60±2.36	78.41±0.06
BotRGCN	89.51±0.99	86.39±1.08	86.94±1.38	86.02±1.46	95.08±0.47	77.46±0.32	46.95±2.10	76.50±1.21	33.95±2.42	64.78±0.93
RGT	89.53±0.30	85.90±0.65	88.20±0.92	84.28±1.19	95.41±0.05	77.82±0.40	39.86±1.77	69.72±1.20	31.22±2.62	72.24±0.38
HGT	87.39±0.41	83.44±0.75	84.47±0.95	82.66±1.24	92.93±0.41	75.66±0.35	33.07±1.57	63.86±2.20	24.02±1.45	60.11±8.74
SimpleHGN	89.33±0.92	86.51±1.06	86.12±1.19	86.94±0.96	95.33±0.46	76.00±0.51	34.95±1.75	63.86±3.05	26.11±1.78	70.05±0.43
GraphMAE	86.51±0.33	82.45±0.51	84.12±0.33	81.20±0.61	92.79±0.45	OOM	OOM	OOM	OOM	OOM
SHAN	86.10±0.30	82.17±0.49	82.49±0.41	81.90±0.84	91.92±0.32	OOM	OOM	OOM	OOM	OOM
SEBot	90.40±0.15	82.06±0.36	81.29±1.28	83.03±1.87	96.01±0.18	OOM	OOM	OOM	OOM	OOM
CACL	-	-	-	-	-	OOT ¹	OOT	OOT	OOT	OOT
BotDGT	-	-	-	-	-	78.37±0.71	53.38±0.93	73.03±0.65	42.06±1.32	79.63±0.38
BotHP	91.10±0.25	82.97±0.57	84.99±0.44	81.06±1.11	95.84±0.14	OOM	OOM	OOM	OOM	OOM
Ours	91.22±0.16	88.57±0.17	88.70±0.34	88.46±0.25	96.08±0.06	78.51±0.12	56.06±1.50	70.50±1.70	46.64±2.51	78.46±0.18

OOM: Out of Memory error; OOT: Out of Time error; - : CACL could not be reproduced, as it relies on raw user data while MGTab only provides processed attributes. BotDGT is not scalable to the dataset.

¹Although we could not reproduce CACL on Twibot-22, the original paper reported Accuracy=75.38 and F1=47.45, which are clearly lower than our results.

masked node features within a graph autoencoding framework.

- **SHAN** [24] models heterogeneous graphs by extracting simplicial complexes and applying hyperbolic attention to fuse multi-order relations.
- **SEBot** [47] is a multi-view graph-based social bot detection method that combines structural entropy minimization with a heterophily-aware encoder, and employs self-supervised contrastive learning to capture node characteristics.
- **CACL** [3] incorporates a community-aware module to mine hard samples and applies graph contrastive learning with multiple graph augmentation strategies.
- **BotDGT** [17] models social networks as dynamic graphs and employs a dynamic graph transformer to incorporate historical context and capture the temporal evolution of user behavior patterns for social bot detection.
- **BotHP** [16] is a generative graph self-supervised framework for social bot detection that combines heterophily-aware representation learning with prototype-guided cluster discovery to model homophilic and heterophilic patterns.

All the competitors have their codes provided by their respective authors, and we adopt the recommended hyperparameters in their papers. Our code is publicly available on GitHub¹.

5.1.3 Implementation. The hyperparameters used in our experiments on the two datasets are summarized in Table 1. We set a cutoff time of 10 days for each method and report the mean and standard deviation over five runs. All experiments are conducted on Linux with NVIDIA V100 GPU (32 GB RAM), 2.50 GHz 40 Intel(R) Xeon(R) Gold 6248 CPU and 376 GB RAM.

5.2 Bot Detection Performance

We evaluate MPS-Bot against a range of representative baselines on two Twitter bot detection benchmarks, with the results summarized in Table 2. Overall, MPS-Bot consistently outperforms the

baselines across most evaluation metrics, demonstrating stronger generalization ability and robustness.

On MGTab, MPS-Bot achieves consistent improvements across all evaluation metrics compared with the best-performing baselines. The gains in Precision, Recall, and F1 demonstrate its ability to enhance recall without sacrificing precision, achieving a more balanced trade-off between the two. This suggests that MPS-Bot not only identifies a broader range of bot accounts but also maintains strong overall discriminative capability.

On TwiBot-22, MPS-Bot delivers superior overall performance, with only a Precision deficit compared to GCN and BotRGCN. The marginally higher Precision of these baselines can be attributed to their reliance on local homophily-based propagation, which, under a highly imbalanced dataset dominated by human accounts, tends to smooth bot features toward the majority class. This smoothing effect reduces false positives, thus increasing Precision, but simultaneously suppresses minority (bot) signals — leading to lower Recall and F1. In contrast, MPS-Bot leverages simplicial complexes to capture higher-order coordination patterns beyond pairwise connections, thereby enhancing detection coverage. Despite the sparsity of simplexes in TwiBot-22 (as analyzed in 5.3), which slightly limits expressiveness and lowers Precision, MPS-Bot maintains competitive Precision and achieves superior overall detection capability.

When compared with baseline methods, several limitations become clear. BotRGCN and RGT incorporate multimodal features or relation graphs for improvement, but they still lag behind MPS-Bot on MGTab and TwiBot-22, especially in F1 and Recall, as they neither explicitly model higher-order structures nor adapt the fusion to user-specific relational preferences. HGT and SimpleHGN utilize heterogeneous graphs to capture diverse relations, yet they overlook simplex-level interactions, limiting their ability to learn comprehensive user representations.

More advanced models such as SEBot, CACL, and BotHP have shown strong results in specific scenarios, yet they also exhibit shortcomings. SEBot and BotHP perform somewhat worse than MPS-Bot on MGTab, suggesting that they may not fully exploit the latent distribution of higher-order features. Moreover, both models

¹<https://github.com/AIandBD/MPS-Bot>

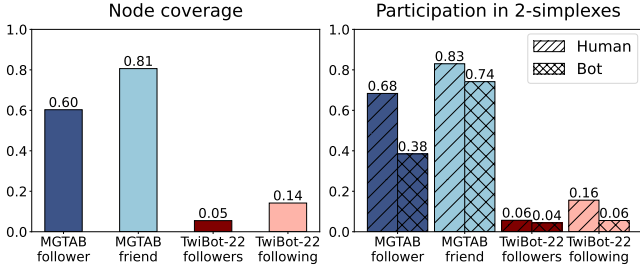


Figure 3: Topology analysis on MGTAB and TwiBot-22.

encounter difficulties on TwiBot-22 due to excessive memory and computational requirements. CACL relies heavily on raw user data, making it unreproducible on MGTAB, and its training on TwiBot-22 did not complete due to excessive runtime. Even considering the original results reported by its authors (Accuracy=75.38, F1=47.45), CACL remains weaker than MPS-Bot.

These results demonstrate that MPS-Bot delivers consistent and significant improvements over existing methods. Its advantage lies in explicitly modeling multi-relational high-order interactions through simplex convolution and effectively integrating them via a user-specific cross-relation dependency attention mechanism. This design enables MPS-Bot to learn robust representations and achieve improved social bot detection performance.

5.3 Topology Analysis

We examine the topological characteristics by analyzing the distribution of 2-simplexes under different relation types in MGTAB and TwiBot-22. The results are presented in Figure 3. Two statistics are considered. Node coverage reflects the proportion of nodes that are involved in at least one 2-simplex, which indicates how widely such higher-order structures are distributed across the graph. Participation in 2-simplexes captures how often human and bot nodes respectively appear in these structures, thereby directly revealing potential class-specific differences that are useful for detection.

In MGTAB, both the follower and friend relations exhibit dense 2-simplex structures, yet they differ in how these structures are organized. The friend relation covers a larger portion of the network, suggesting that clustering patterns are widely shared across nodes. In contrast, the follower relation, while involving fewer nodes overall, shows a much clearer separation between human and bot participation. This implies that even in a dense network, different relations emphasize distinct aspects of the topology: the friend relation highlights broad clustering, whereas the follower relation provides stronger class-discriminative structural cues.

In TwiBot-22, the overall graph is much sparser, but the contrast between relations is even sharper. The following relation exhibits substantially higher node coverage and shows a clear participation gap between humans and bots, indicating that it contains most of the useful higher-order structures. By comparison, the followers relation is extremely sparse, covering very few nodes and showing almost no difference between classes. This suggests that in TwiBot-22, the high-order structural information is concentrated primarily in the following relation, compared with the followers relation.

These observations demonstrate that the distribution of simplexes varies not only across datasets but also between humans and bots under different relations.

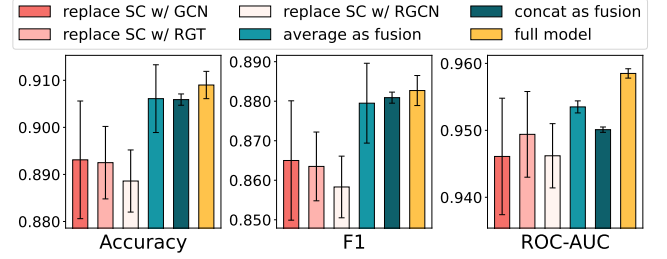


Figure 4: Performance (Accuracy, F1-score, and ROC-AUC) with standard deviations in the ablation study on MGTAB. SC refers to the SimplexConv layer.

5.4 Ablation Study

We conduct ablation studies to evaluate the contribution of each component in MPS-Bot. Specifically, we remove or replace individual modules and report the results on MGTAB in Figure 4.

For the higher-order modeling module, we replace the proposed simplex convolutional layer with alternative methods that operate only on non-higher-order structures, including GCN, RGT, and RGCN. We observe a clear drop in all three metrics (Accuracy, F1, and ROC-AUC) across these variants. This performance degradation demonstrates that message passing over simplexes plays a crucial role in capturing the underlying topological structures of social networks. Without such higher-order representations, the model struggles to effectively capture the complex connectivity patterns.

For the relation fusion module, we compare our user-specific cross-relation dependency attention mechanism with two alternative fusion strategies: simple averaging and concatenation. Both variants lead to a noticeable decline in performance compared to the full MPS-Bot model. This highlights that a naive fusion strategy is insufficient, as it either ignores the dependencies between relations (in the case of averaging) or introduces redundant and noisy signals (in the case of concatenation). By contrast, MPS-Bot adaptively models relation dependencies and tailors fusion to each user.

In summary, the ablation study demonstrates that both the simplex convolutional layer for higher-order modeling and the user-specific cross-relation dependency attention mechanism for adaptive relation fusion are crucial to MPS-Bot’s effectiveness.

5.5 Sensitivity Analysis

We conduct a sensitivity analysis of the hyperparameters dropout and learning rate on MGTAB, with the results shown in Figure 5. Specifically, we vary dropout from 0.1 to 0.5 in increments of 0.1, and the learning rate from 1×10^{-4} to 1×10^{-2} .

The results reveal clear trends across the three evaluation metrics. Increasing the learning rate generally improves performance at first, but beyond a certain threshold, further increases lead to performance degradation. This is likely because an excessively large learning rate makes the optimization unstable, preventing the model from converging to a good local optimum. In contrast, raising the dropout rate consistently reduces performance, while lower dropout values yield better results. A possible reason is that excessive dropout disrupts the representation learning process by discarding too much structural information, which is crucial for modeling user interactions in social networks.

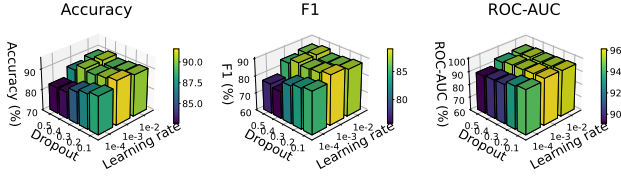


Figure 5: Sensitivity analysis of the hyperparameters dropout and learning rate on MG TAB, illustrating their effects on model performance (Accuracy, F1-score, and ROC-AUC).

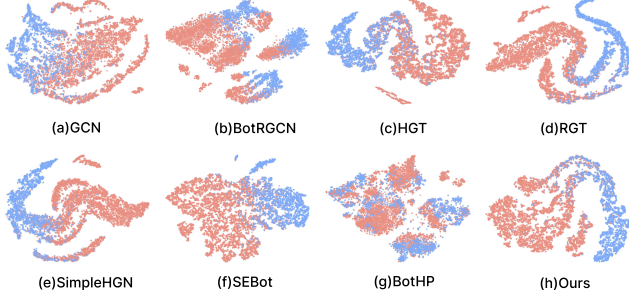


Figure 6: Visualization of user node representations on MG TAB. Red represents humans, and blue represents bots.

These observations suggest that balancing dropout and learning rate is critical to avoid underfitting or overfitting and ensure stable and effective training.

5.6 Visualization

We visualize the learned user node representations using *t*-SNE [31] to compare MPS-Bot with several baselines on MG TAB, as shown in Figure 6. Red and blue denote humans and bots, respectively. Overall, MPS-Bot produces more compact and clearly separated clusters than other methods, demonstrating stronger representation and discrimination ability.

For the baselines, GCN representations show heavy overlap between classes, reflecting its limitation in modeling complex relations. BotRGCN and RGT achieve moderate separation but the clusters remain scattered, suggesting that multimodal or relational modeling alone is insufficient. HGT and SimpleHGN provide clearer structures, yet class boundaries are still blurred. SEBot captures localized community patterns, but the representations lack global smoothness. BotHP yields relatively distinct groups, though mixing across classes is still evident.

By contrast, MPS-Bot generates representations that are both compact and well separated by class boundaries. This advantage stems from the simplex convolutional layer, which enriches node representations with higher-order structural information, and the user-specific cross-relation dependency attention mechanism, which adaptively fuses relation-specific representations according to each user's interaction patterns. These components together enable MPS-Bot to learn more expressive and robust representations, leading to stronger performance in social bot detection.

5.7 Case Study

As shown in Figure 7, we analyze a well-disguised bot account that appears human-like in its surface features. Metadata enhance

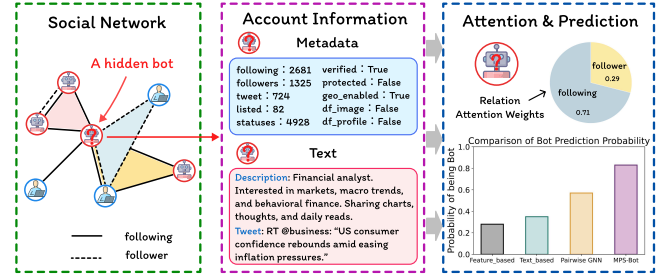


Figure 7: Case study of a well-disguised bot account. Although the account appears human-like in metadata and text, MPS-Bot uncovers hidden coordination through simplexes and adaptively emphasizes the following relation reflecting its user-specific preference, leading to a more accurate bot classification.

its overall credibility, while the profile description closely resembles that of a genuine user, and its tweets share financial news in a natural tone, making it challenging for feature-based or text-based detectors to identify. However, MPS-Bot captures hidden coordination patterns that are not easily observable from pairwise links. Through message passing over simplexes, the model uncovers higher-order group behaviors that subtly indicate coordinated automation. Moreover, the user-specific cross-relation dependency modeling assigns a higher attention weight to the following relation, indicating that the model learns richer relational information from this relation. By adaptively emphasizing this more informative relation, MPS-Bot achieves more accurate bot classification.

6 Conclusion

In this paper, we present MPS-Bot, a practical model for bot detection that introduces higher-order interaction into social network modeling through message passing over simplexes. By exploiting simplexes derived from the network, the model enriches node representations with group coordination patterns beyond pairwise interactions. Furthermore, a user-specific cross-relation dependency modeling mechanism is employed to adaptively integrate relation-specific signals in alignment with each user's relational preferences, enabling more discriminative multi-relation fusion.

We evaluate MPS-Bot against a broad range of state-of-the-art methods on two widely used Twitter bot detection benchmarks. The extensive experimental results demonstrate that MPS-Bot generally outperforms competitive baselines, validating the effectiveness of combining higher-order structural signals with user-specific relational dependencies. These findings highlight the potential of higher-dimensional message passing over simplexes as a powerful approach for unmasking bots in social networks.

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